

Subsets Cardinality and Types of Sets

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A **set** is a well-defined collection of objects called **elements**.

Subset

A **subset** is a set whose every element belongs to another set.

If A is a subset of B , we write

$$A \subseteq B$$

Empty Set

The **empty set**, denoted by $\emptyset$, is a subset of every set.

$$\emptyset \subseteq A$$

for every set A .

Every Set Is a Subset

Every set is a subset of itself.

$$A \subseteq A$$

For example, if

$$A = \{1, 2, 3\},$$

then $A \subseteq A$.

Cardinality

The **cardinality** of a set is the number of elements it contains.

It is written as $n(A)$ or $|A|$.

$$A = \{2, 4, 6, 8\}$$

Therefore,

$$n(A) = |A| = 4$$

Finite Set

A **finite set** has a limited number of elements.

$$A = \{5, 10, 15, 20\}$$

The set A has 4 elements, so it is finite.

Infinite Set

An **infinite set** has endlessly many elements.

$$B = \{1, 2, 3, \dots\}$$

The set B is infinite.

Number of Subsets

A set with n elements has 2^n subsets.

For example, a set with 3 elements has

$$2^3 = 8$$

subsets.

Examples

Example 1

List all the subsets of $A = \{1, 2\}$.

$\emptyset, \quad \{1\}, \quad \{2\}, \quad \{1, 2\}$

There are $4 = 2^2$ subsets.

Example 2

Determine the cardinality of

$$A = \{5, 10, 15, 20\}.$$

There are 4 elements.

$$n(A) = |A| = 4$$

Example 3

Tell whether the set is finite or infinite.

$$A = \{1, 2, 3, \dots\}$$

The set has endlessly many elements.

Therefore, A is **infinite**.

Example 4

Let $D = \{a, b, c\}$.

Is $\{a, c\}$ a subset of D ?

Yes, because every element of $\{a, c\}$ belongs to D .

$$\{a, c\} \subseteq D$$

Example 5

Without listing all the subsets, determine the number of subsets of

$$A = \{a, b, c\}.$$

Since A has 3 elements,

$$2^3 = 8$$

Therefore, A has 8 subsets.

Practice Exercises

Practice 1

List all the subsets of

$$A = \{1, 2\}.$$

Practice 2

List all the subsets of

$$B = \{a, b, c\}.$$

Practice 3

List all the subsets of

$$C = \{x, y, z, w\}.$$

Practice 4

Determine the cardinality of each set.

$$(a) \{5, 10, 15, 20\}$$

$$(b) \{a, e, i, o, u\}$$

$$(c) \emptyset$$

Practice 5

Determine the cardinality of each set.

$$(a) \{2, 4, 6, 8, 10, 12\}$$

$$(b) \{m, n, p\}$$

$$(c) \{100\}$$

Practice 6

Tell whether each set is finite or infinite.

(a) $\{1, 2, 3, \dots\}$

(b) $\{5, 10, 15\}$

(c) The set of vowels in the English alphabet.

Practice 7

Tell whether each set is finite or infinite.

(a) The set of all whole numbers.

(b) The set of months in a year.

(c) The set of even numbers.

Practice 8

Let $D = \{a, b, c\}$.

(a) Is $\{a, c\}$ a subset of D ?

(b) Is $\{a, d\}$ a subset of D ?

(c) Is $\emptyset$ a subset of D ?

Practice 9

Let $E = \{2, 4, 6, 8\}$.

(a) Is $\{4, 8\}$ a subset of E ?

(b) Is $\{2, 5\}$ a subset of E ?

(c) Is E a subset of itself?

Practice 10

Without listing all the subsets, determine the number of subsets of each set.

$$(a) \{a, b\}$$

$$(b) \{1, 2, 3\}$$

$$(c) \{m, n, p, q\}$$

Activities

Activity 1

List all the subsets of

$$A = \{3, 5\}.$$

Activity 2

List all the subsets of

$$B = \{p, q, r\}.$$

Activity 3

List all the subsets of

$$C = \{2, 4, 6\}.$$

Activity 4

Determine the cardinality of each set.

$$(a) \{7, 14, 21\}$$

$$(b) \{b, c, d, e, f, g\}$$

$$(c) \emptyset$$

Activity 5

Determine the cardinality of each set.

$$(a) \{11, 22, 33, 44, 55\}$$

$$(b) \{x, y\}$$

$$(c) \{50\}$$

Activity 6

Tell whether each set is finite or infinite.

(a) $\{2, 4, 6, 8, \dots\}$

(b) $\{8, 16, 24, 32\}$

(c) The set of days in a week.

Activity 7

Tell whether each set is finite or infinite.

(a) The set of integers.

(b) The set of planets in the Solar System.

(c) The set of odd numbers.

Activity 8

Let $D = \{m, n, o\}$.

(a) Is $\{m, o\}$ a subset of D ?

(b) Is $\{m, p\}$ a subset of D ?

(c) Is $\emptyset$ a subset of D ?

Activity 9

Let $E = \{10, 20, 30, 40\}$.

(a) Is $\{20, 40\}$ a subset of E ?

(b) Is $\{10, 25\}$ a subset of E ?

(c) Is E a subset of itself?

Activity 10

Without listing all the subsets, determine the number of subsets of each set.

(a) $\{x, y\}$

(b) $\{4, 5, 6\}$

(c) $\{a, b, c, d\}$

Thank you!